Assignment

You are a security analyst working for a financial institution, and you've been tasked with securing sensitive transaction data that will be stored in the company's internal database. Since this data must be secure, you'll use symmetric encryption to ensure it remains confidential (scenario shown in figure 1). Your task is to create a small, secure encryption and decryption module as metioned in the requirement section.

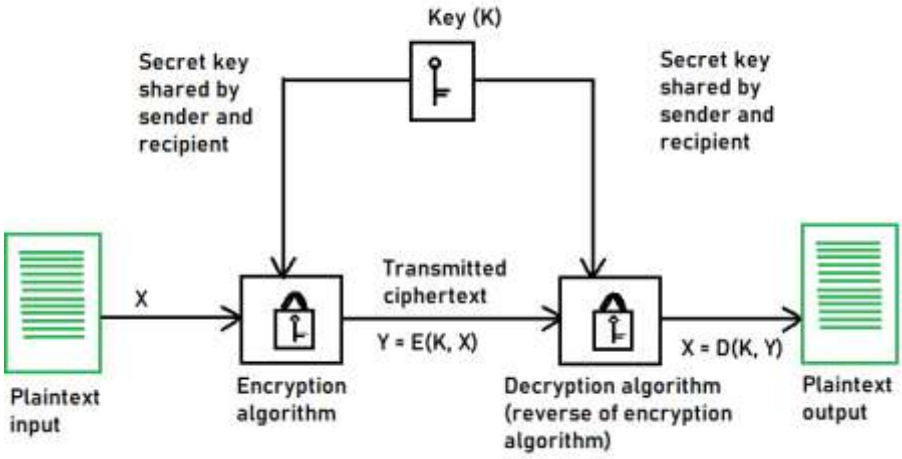

Figure 1: Symmetric key cryptography

Requirements

1. **Encrypt and Decrypt Transaction Data**: Write code to encrypt and decrypt transaction data using:
    1. **Caesar Cipher:** The Caesar Cipher is one of the simplest and oldest methods of encrypting messages, named after Julius Caesar, who reportedly used it to protect his military communications. This technique involves shifting the letters of the alphabet by a fixed number of places. For example, with a shift of three, the letter 'A' becomes 'D', 'B' becomes 'E', and so on. Despite its simplicity, the Caesar Cipher formed the groundwork for modern cryptographic techniques.

        Cryptography Algorithm For the Caesar Cipher

        To cipher a given text we need an integer value, known as a shift which indicates the number of positions each letter of the text has been moved down.

The encryption can be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, A = 0, B = 1,…, Z = 25. Encryption of a letter by a shift n can be described mathematically as.

For example, if the shift is 3, then the letter A would be replaced by the letter D, B would become E, C would become F, and so on. The alphabet is wrapped around so that after Z, it starts back at A.

Here is an example of how to use the Caesar cipher to encrypt the message "HELLO" with a shift of 3:

1. Write down the plaintext message: HELLO
2. Choose a shift value. In this case, we will use a shift of 3.
3. Replace each letter in the plaintext message with the letter that is three positions to the right in the alphabet.

   H becomes K (shift 3 from H)
   E becomes H (shift 3 from E)
   L becomes O (shift 3 from L)
   L becomes O (shift 3 from L)
   O becomes R (shift 3 from O)

4. The encrypted message is now "KHOOR".

To decrypt the message, you simply need to shift each letter back by the same number of positions. In this case, you would shift each letter in "KHOOR" back by 3 positions to get the original message, "HELLO".

Examples:

```
Text : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Shift: 23
Cipher: XYZABCDEFGHIJKLMNOPQRSTUVW

Text : ATTACKATONCE
Shift: 4
Cipher: EXXEGOEXSRGI
```

2. **Playfair Ciphers:** The Playfair cipher is a manual encryption technique and a significant milestone in cryptographic history. This cipher encrypts digraphs, or pairs of letters, rather than individual characters. This method provides a higher security level than simple substitution ciphers by reducing character frequency analysis attacks, as each letter pair is encrypted uniquely. The encryption process uses a 5x5 grid containing a keyword, where each letter of the alphabet (except 'J', which is typically combined with 'I') is placed within the matrix. To encrypt, pairs of letters from the plaintext are identified, and specific rules based on their positions in the matrix are applied: if the letters are in the same row or column, each letter is replaced by the next one in that row or column; if the letters form a rectangle, they are substituted by the letters on the opposite corners of the rectangle. This approach results in an encrypted text that is difficult to decipher without knowing the keyword. The Playfair cipher, while innovative, has limitations: it is vulnerable to frequency analysis of digraphs in longer texts and is less secure than modern cryptographic methods. However, the Playfair cipher's contribution to cryptography is notable for its simplicity, ease of manual implementation, and historical significance. Its underlying principles are foundational for understanding the evolution of encryption techniques and shifting from monoalphabetic to polygraphic ciphers.
3. **Hill Cipher:** Hill cipher is a polygraphic substitution cipher based on linear algebra. Each letter is represented by a number modulo 26. The simple scheme A = 0, B = 1, …, Z = 25 is often used, but this is not an essential cipher feature. Each block of n letters (considered as an n-component vector) is multiplied by an invertible n × n matrix to encrypt a message, against modulus 26. To decrypt the message, each block is multiplied by the inverse of the matrix used for encryption. The matrix used for encryption is the cipher key, and it should be chosen randomly from the set of invertible n × n matrices (modulo 26).

Examples:

```
Input  : Plaintext: ACT
         Key: GYBNQKURP
Output : Ciphertext: POH
```

```
Input  : Plaintext: GFG
         Key: HILLMAGIC
Output : Ciphertext: SWK
```

Encryption

We have to encrypt the message ‘ACT’ (n=3).The key is ‘GYBNQKURP’ which can be written as the n×n matrix:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

The message ‘ACT’ is written as a vector:

$$\begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$

The encrypted vector is given as:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix} \equiv \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \pmod{26}$$

which corresponds to ciphertext of ‘POH’

Decryption

To decrypt the message, we turn the ciphertext back into a vector, then simply multiply by the inverse matrix of the key matrix (IFKVIVVMI in letters). The inverse of the matrix used in the previous example is:

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}^{-1} \equiv \begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \pmod{26}$$

For the previous Ciphertext ‘POH’:

$$\begin{bmatrix} 8 & 5 & 10 \\ 21 & 8 & 21 \\ 21 & 12 & 8 \end{bmatrix} \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} \equiv \begin{bmatrix} 260 \\ 574 \\ 539 \end{bmatrix} \equiv \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} \pmod{26}$$

which gives us back ‘ACT’.

2. **Save and Load Keys**: Write functions to save the encryption key to a secure file (.txt) and to load it back when needed. (Take encryption key as input from console and save it in a text file).

3. **Encrypt and Decrypt Functionality**: Create functions that accept plaintext from a (.txt) file for encryption and ciphertext (.txt) for decryption, demonstrating successful encryption and decryption of data.
4. **Measure the time:** Measure the time it takes to encrypt and decrypt the text using these algorithms **(Caesar Cipher, Playfair cipher, and Hill Cipher)**. Assume keys as per your choice. You can use programming languages like Python, Java, or C++ and their respective time libraries to record the encryption and decryption time.
5. Identify Which algorithm took less time for encryption?
6. Identify Which algorithm took less time for decryption?
7. How do these results reflect real-world use cases where encryption speed and security must be balanced?

You have to upload a single Word document (along with a video) having the following components:

1. **Codes for the mentioned algorithms.**
2. **Your encryption keys for the algorithms.**
3. **Generated ciphertexts by each algorithm.**
4. **Graphs having encryption time and decryption time.**
5. **What is your understanding of how these results reflect real-world use cases where encryption speed and security must be balanced?**